

Deep Learning for Audio

Lecture 11

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AI Masters

2026

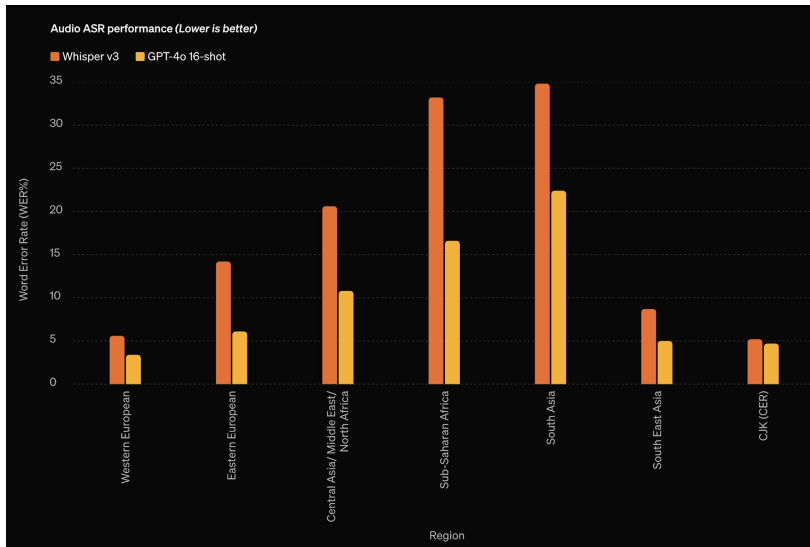
Outline

1. LLM-based audio models overview
2. Audio Conditioned LLMs: Seed-ASR, Llama3, Phi4, Voxtral
3. LLM-based TTS: TTS-1, FishSpeech, Qwen3-TTS
4. End-to-end speech interaction LLMs: SpeechGPT, Mini-Omni, Llama-Omni, Moshi, Qwen3-Omni

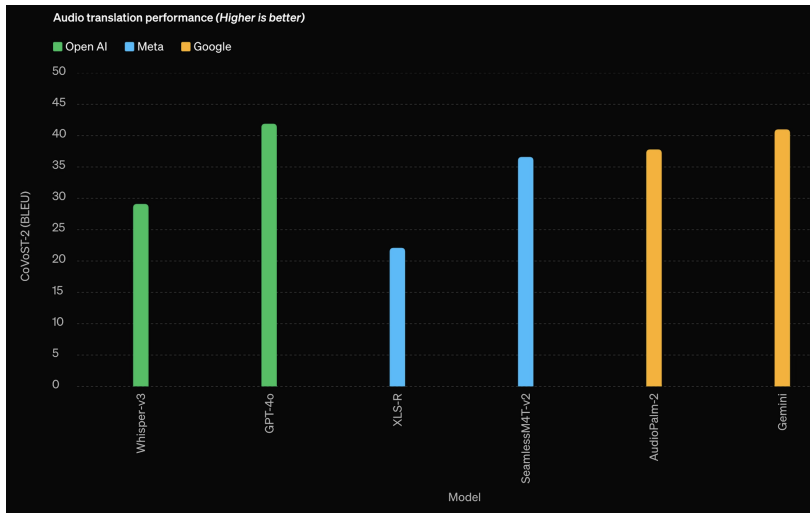
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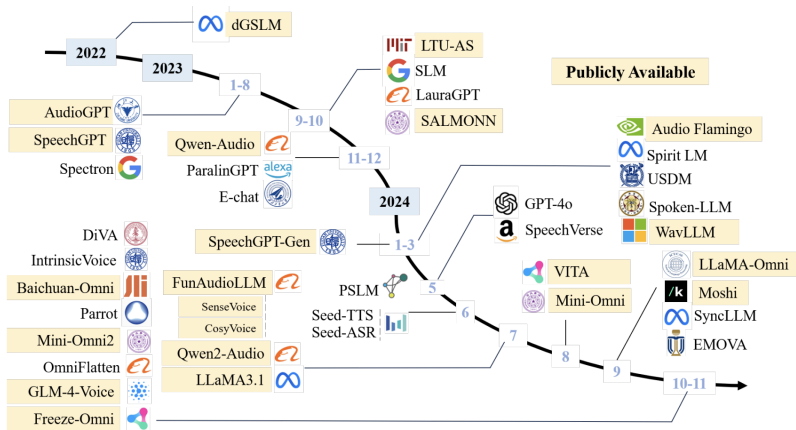
Effectiveness of Spoken LLMs



Effectiveness of Spoken LLMs



Timeline of spoken dialogue models in recent years (2022-2024)



Publicly available model checkpoints: yellow color

Metrics: comes from tasks we want SLM to solve



Hi, can you give me some travel tips for Japan?



Absolutely! Japan is a fantastic destination with a wonderful mix of traditional culture and modern attractions. If you're heading to Tokyo, make sure to visit the historic Senso-ji Temple and experience the lively atmosphere at Shibuya Crossing. Don't miss out on the incredible views from Tokyo Tower and be sure to try some authentic sushi and ramen while you're there. Kyoto is perfect for immersing yourself in traditional Japanese culture, with beautiful sites like the Golden Pavilion and Kiyomizu-den Temple. You can also enjoy a serene tea ceremony in one of the local tea houses. If you're a foodie, Osaka is a must-visit with its delicious takoyaki and okonomiyaki...

(a) Text Intelligence



I've been feeling down lately. Can you talk with me?



(sad tone): I've been feeling down lately. Can you talk with me?



(gentle, encouraging tone, gentle mezzo-soprano): Of course, I'm here for you. I'm sorry to hear you're feeling this way. Remember, you're strong enough to overcome them. Anything specific on your mind that I can help with?



Can you tell me a joke in a humorous tenor voice?



(humorous tone, tenor): Sure thing! Here goes: How do trees get on the Internet? They log in!

(b) Speech Intelligence



I love the sound of birds chirping. Can you play that for me?



Certainly, here's a relaxing bird song for you.



That's perfect! Now, could you play some soothing piano music?



Sure thing, here's a gentle piano melody.



(Playing calming piano music)

(c) Audio and Music Generation



Hey, I just recorded an audio of my dog chasing his tail and barking excitedly. Can you write a funny poem based on it?



(An audio of a dog chasing his tail and barking excitedly)



Sounds like a lot of fun! Here's a playful poem inspired by your energetic pup:

(lyrical, humor tone):

*Round and round he spins with glee,
A fluffy tornado, wild and free,
With barks that echo, loud and clear,
He's the happiest pup far and near!
His tail's his goal, he chases fast,
In his mind, he'll catch it at last.
But round he goes, a furry spree,
A doggy dance, pure jubilee!*

(d) Audio and Music Understanding



Hi, could you translate this into French for me? "The weather is beautiful today, and I'd like to go for a walk in the park."



Absolutely! The translation in French is: "Il fait beau aujourd'hui, et j'aimerais aller me promener dans le parc."



Thanks! Now translate this to Spanish: "I'm very happy to meet you."



In Spanish, that would be: "Estoy muy feliz de conocerte."

(e) Multilingual Capability



So far, you have a pair of running shoes, a water bottle, and a set of wireless earbuds in your shopping list. Need anything else?



Can you add milk and bread to my shopping list?



Here's the updated list: you've got running shoes, a water bottle, wireless earbuds, milk, and bread. Anything else you'd like to add, or are we all set?



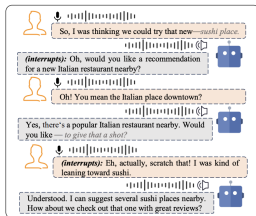
Actually, let's throw in a box of granola bars too.



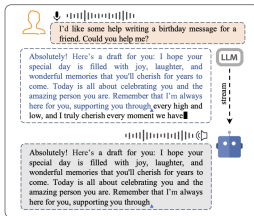
Got it—adding a box of granola bars. So now your cart has running shoes, a water bottle, wireless earbuds, milk, bread, and granola bars. Looking good!

(f) Context Learning

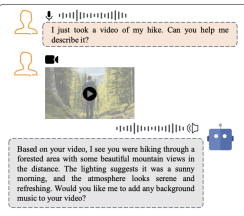
Metrics: comes from tasks we want SLM to solve



(g) Interaction Capability



(h) Streaming Latency



(i) Multimodal Capability

LLM training stages: where to add audio modality?

1. Pre-training
 - ▶ SSL, next token prediction
 - ▶ the most expensive part, usually done only on text
2. → Supervised fine-tuning (SFT, training on instructive data)
3. RL (RLHF, DPO, GRPO, etc.)
 - ▶ further improving using human feedback
 - ▶ hard to implement but adds quality improvements

LLM+Speech Input Training Data: example from Qwen-Audio

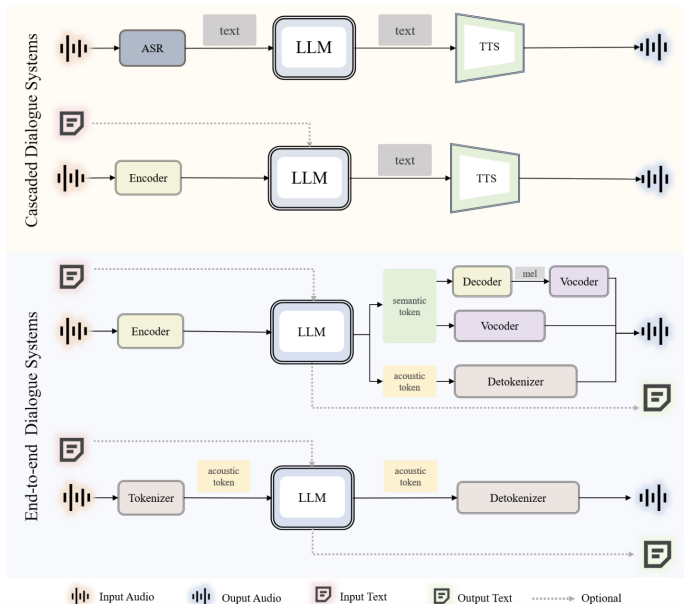
Types	Task	Description	Hours
Speech	ASR	Automatic speech recognition (multiple languages)	30k
	S2TT	Speech-to-text translation	3.7k
	OSR	Overlapped speech recognition	<1k
	Dialect ASR	Automatic dialect speech recognition	2k
	SRWT	English speech recognition with word-level timestamps	10k
		Mandarin speech recognition with word-level timestamps	11k
	DID	Dialect identification	2k
	LID	Spoken language identification	11.7k
	SGC	Speaker gender recognition (biologically)	4.8k
	ER	Emotion recognition	<1k
	SV	Speaker verification	1.2k
	SD	Speaker diarization	<1k
	SER	Speech entity recognition	<1k
	KS	Keyword spotting	<1k
	IC	Intent classification	<1k
	SF	Slot filling	<1k
Sound	SAP	Speaker age prediction	4.8k
	VSC	Vocal sound classification	<1k
	AAC	Automatic audio caption	8.4k
	SEC	Sound event classification	5.4k
	ASC	Acoustic scene classification	<1k
Music&Song	SED	Sound event detection with timestamps	<1k
	AQA	Audio question answering	<1k
	SID	Singer identification	<1k
	SMER	Singer and music emotion recognition	<1k
	MC	Music caption	25k
	MIC	Music instruments classification	<1k
	MNA	Music note analysis such as pitch, velocity	<1k
	MGR	Music genre recognition	9.5k
	MR	Music recognition	<1k
	MQA	Music question answering	<1k

Chu et al., Qwen-Audio: Advancing Universal Audio Understanding via Unified Large-Scale Audio-Language Models, Alibaba, 2023

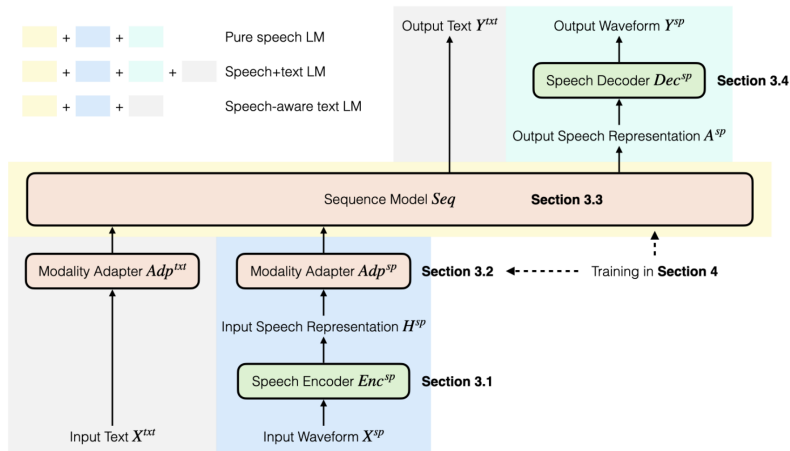
LLM+Speech Input Training Data: format

1. **System prompt** ("You are a helpful assistant. Do as I say.")
2. Optional: **Context** (previous discussion between the user and the assistant)
3. **Task prompt** ("Write down the language of the audio clip and transcribe it.")
4. **Audio embeddings**, projected into the LLM space
5. **Target answer** – we learn this

General overview of current spoken dialogue systems



Speech Language Model (SLM) architecture



Audio encoder: Comparison of semantic and acoustic audio representations

	Advantages of the comprehension side	Performance of unify music and audio	Compression rate of speech	Emotional and acoustic information	Pipeline for post-processing
Semantic	Strong	Weak	High	Less	Cascade
Acoustic	Weak	Strong	Low	More	End-to-end

- ▶ **Semantic representations:** usually pre-trained in a self-supervised fashion or on a downstream task
 - ▶ Extract the high-level features, allowing us to distill the most important aspects from the audio bit
 - ▶ **HuBERT, USM, wav2vec 2.0, Whisper encoder**
 - ▶ If you don't need audio output, semantic representations are probably more efficient
- ▶ **Acoustic representations:** obtained via reconstruction train
 - ▶ These preserve a lot more additional info as to how the words were pronounced, which background noise was present, etc.
 - ▶ **Encodec, WavTokenizer, SoundStream** (uses discrete representations)

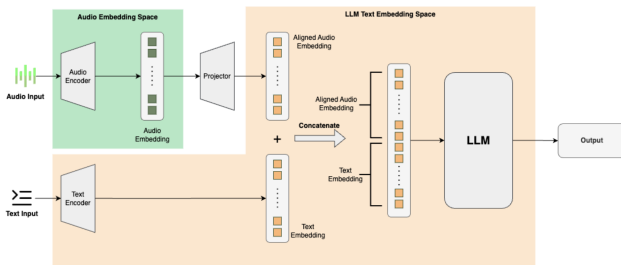
Modality Adaptor: why do we need it

Typical Modality Adapter: Subsample + Linear Projection

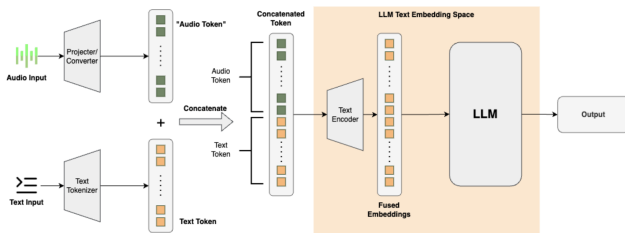
Why use:

- ▶ **Text and audio embeddings may have incompatible dimensions**
 - ▶ Text: $(\text{length in tokens}) \times (\text{emb_dim})$
 - ▶ Audio: $(\text{audio length} / \text{stride}) \times (\text{audio_enc_dim})$
- ▶ **Audio embeddings may have a large token rate:**
 - ▶ Spectrograms usually have a 10 ms stride $\rightarrow$ 100 tokens/sec
 $\rightarrow$ we need context of 360k to fit an hour of audio
 - ▶ Most LLMs don't have that much. Do we really need that much?

Continuous vs Discrete audio input



(a) Continuous Embedding Projection

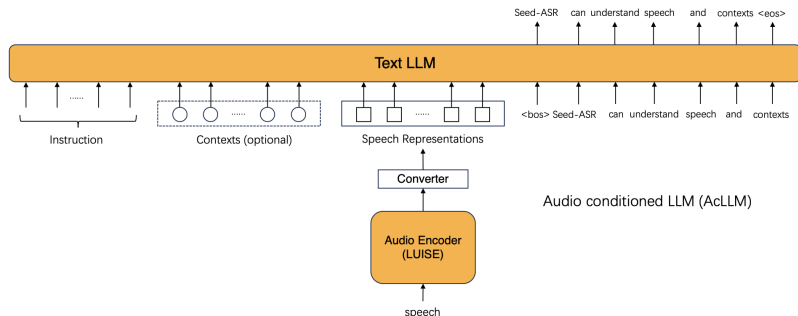


(b) Token Mapping

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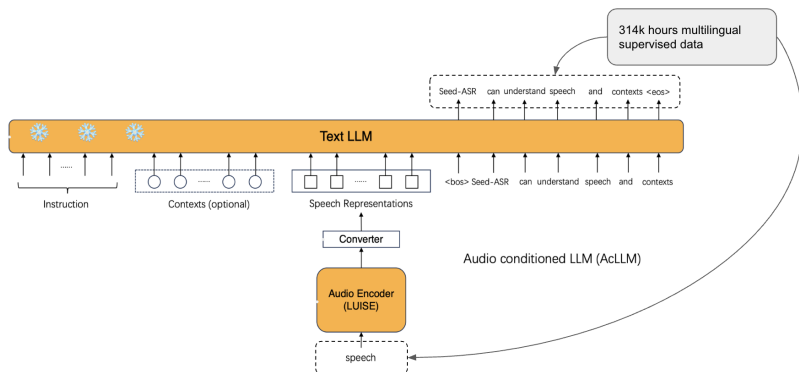
Seed-ASR



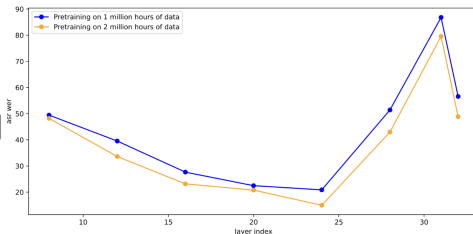
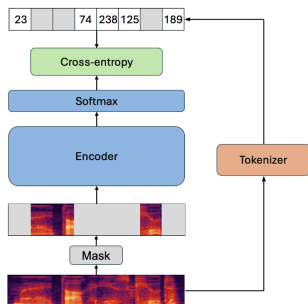
- ▶ Text LLM: Mixture of Experts LLM with tens of billions parameters
- ▶ Audio Encoder: 12.4M hours unlabeled multilingual data, 2B parameters
- ▶ Converter: sequence reduction + projection, 1 frame = 160ms

Seed Team, Seed-ASR: Understanding Diverse Speech and Contexts with LLM-based Speech Recognition, ByteDance, 2024

Seed-ASR: SFT finetuning



Seed-ASR Ecnoder



- ▶ Two stage training
- ▶ BEST-RQ → probing → K-Means vocab

Seed-ASR Scaling Laws

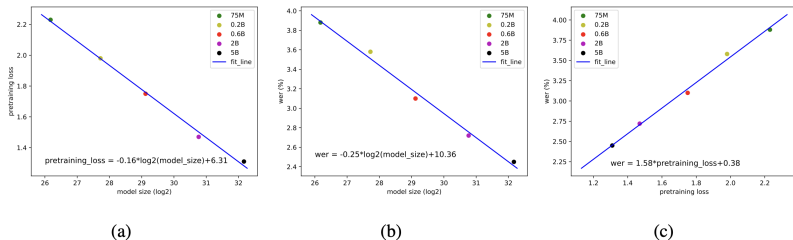


Figure: (a) depicts the correlation between the pretraining loss of our audio encoder (LUISE) and base-2 logarithm of the model parameter size. (b) depicts the correlation between the greedy WER after the SFT and base-2 logarithm of the model parameter size. (c) depicts the correlation between the greedy WER after SFT and the pretraining loss of LUISE

Seed-ASR: training



Model	Multidomain WER ↓	Hardcase (F1%) ↑	Context Strict (Recall%) ↑
Model after Context SFT	2.02	93.39	80.63
+ RL w/ WER reward	1.98	93.39	75.34
+ RL w/ Weighted WER reward	1.94	93.78	78.01
+ train w/ context	1.94	93.72	80.63

Seed-ASR: results

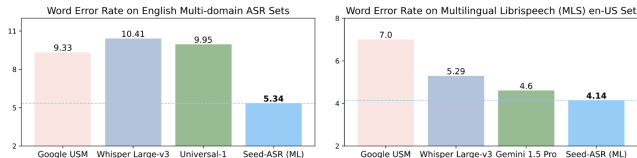


Table 10: Comparison with Google USM, Whisper Large v3 and Universal-1 on English multi-domain, multi-accent, hardcase evaluation sets, and multilingual multi-domain evaluation sets.

	Google USM[50]	Whisper Large v3[39]	Universal-1[41]	Seed-ASR (ML)
English Multi-domain (WER%) ↓	9.33	10.41	9.95	5.34
English Multi-accent (WER%) ↓	22.19	21.52	14.40	11.26
English Hardcase (F1%) ↑	63.30	79.54	77.82	87.94
Multilingual Multi-domain (WER%) ↓	21.51	20.55	-	12.16

Llama3.1 & speech



Figure: Architecture of our speech interface for Llama 3

- ▶ **Speech Adapter:** 100M, Convolution, rotary Transformer, Linear; stride=80ms
- ▶ **Speech Encoder:** Conformer-1B, 15M hours BEST-RQ pre-train, stride=40ms, $kernel_{size} = 7$, rotary attention

Llama3.1 & speech: speech encoder SFT

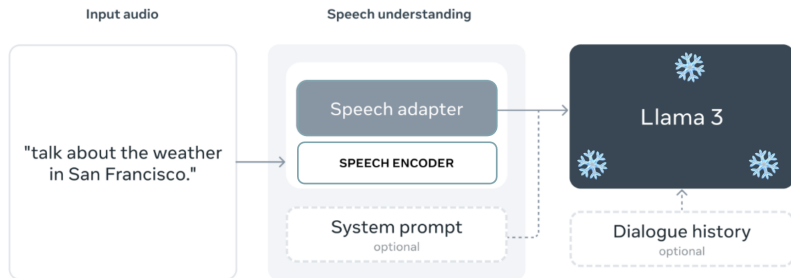


Figure: Freeze LLM Layers and finetune speech encoder

Supervised Fine-Tuning Data:

- ▶ 230k hours ASR (34 lang)
- ▶ 90k hours AST ($X \rightarrow \text{En}$; $\text{En} \rightarrow X$)
- ▶ **Dialogue Data:** 60k hours AudioChatLlama-like + 25k hours text SFT + TTS

Llama3.1 & speech: AudioChatLlama

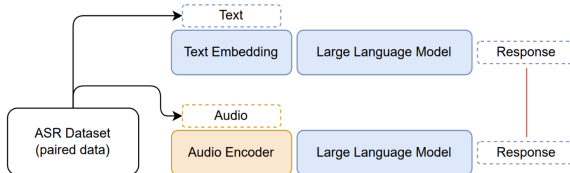


Figure 2: The aim is to create an end-to-end system that would generate the same response when being fed spoken (audio) input instead of its text version. The overall system should be invariant to the modality of the inputs containing the same semantic information.

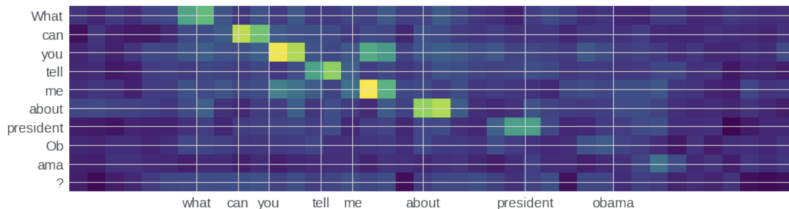


Figure 3: Cosine similarity between text and audio embeddings.

Llama3.1 & speech: results

	Llama 3 8B	Llama 3 70B	Whisper	SeamlessM4T v2	Gemini 1.0 Ultra	Gemini 1.5 Pro
MLS (English)	4.9	4.4	6.2 (v2)	6.5	4.4	4.2
LibriSpeech (test-other)	3.4	3.1	4.9 (v2)	6.2	—	—
VoxPopuli (English)	6.2	5.7	7.0 (v2)	7.0	—	—
FLEURS (34 languages)	9.6	8.2	14.4 (v3)	11.7	—	—

Table 31 Word error rate of our speech interface for Llama 3 on speech recognition tasks. We report the performance of Whisper, SeamlessM4T, and Gemini for reference.

	Llama 3 8B	Llama 3 70B	Whisper v2	SeamlessM4T v2
FLEURS (33 lang. → English)	29.5	33.7	21.9	28.6
Covost 2 (15 lang. → English)	34.4	38.8	33.8	37.9

Table 32 BLEU score of our speech interface for Llama 3 on speech translation tasks. We report the performance of Whisper and SeamlessM4T for reference.

audio conversation

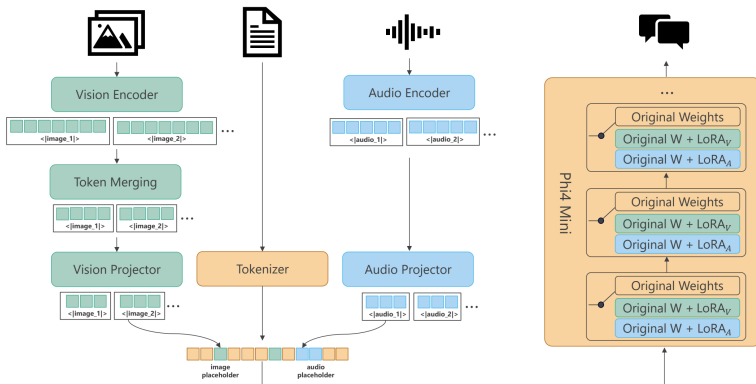
Hey, do you know the movie "Ein Fisch namens Wanda"? Sorry, I don't know the English title

I think I can help you out! I believe the movie you're thinking of is called "A Fish Called Wanda" in English. It's a 1988 British comedy film directed by Charles Crichton, written by John Cleese and Charles [...]

So, at some point, John Cleese says "Le due cupole grandi della cattedrale di Milano." Was heisst'n das eigentlich?

"Le due cupole" ist italienisch und bedeutet "die zwei Kuppeln". Die Kathedrale von Mailand, auch Duomo von Mailand genannt, ist ein berühmtes Bauwerk in Mailand, Italien, und es hat tatsächlich zwei Kuppeln: die Kuppel des Hauptbaus und die Kuppel des Campanile (Glockenturms) [...]

Phi4 (Microsoft)



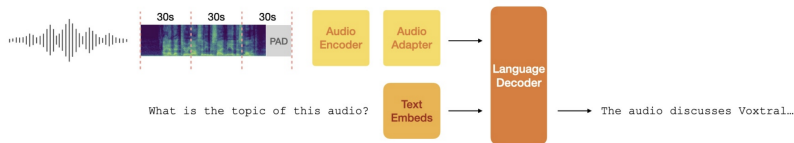
Audio encoder: 3 conv layers + 24 conformer blocks, **Audio projector:** 2-layer MLP (460M params total)

Phi-4-Mini Technical Report: Compact yet Powerful Multimodal Language Models via Mixture-of-LoRAs, Microsoft, 2025

Phi4: audio results

Task	Metric	Dataset	Phi-4-Multimodal 5.6B	WhisperV3 1.5B	SeamlessM4T-V2 2.3B	Qwen2-audio 8B	Gemini- 2.0-Flash	GPT-4o -
ASR	WER ↓	CV15	6.80	8.13	8.46	8.55	9.29	18.14
		FLEURS	4.00	4.58	7.34	8.28	4.73	5.42
		OpenASR	6.14	7.44	20.70	7.43	8.56	15.76
AST	BLEU ↑	Inference Type	(0-shot, CoT)	0-shot	0-shot	0-shot	0-shot	0-shot
		CoVoST2 X-EN	(39.33, 40.76)	33.26	37.54	34.80	36.62	37.09
		CoVoST2 EN-X	(37.82, 38.73)	N/A	32.84	34.04	35.93	37.19
		FLEURS X-EN	(29.86, 32.35)	25.76	28.87	23.72	30.69	32.61
		FLEURS EN-X	(32.15, 33.56)	N/A	30.44	23.24	37.33	36.78
SQQA	Score 1-10 ↑	MT-Bench	7.05	N/A	N/A	4.92	8.07	8.11
	ACC ↑	MMMLU	38.50	N/A	N/A	15.53	72.31	72.56
SSUM	Score 1-7 ↑	Golden3	6.28	N/A	N/A	2.25	6.29	6.76
		AMI	6.29	N/A	N/A	1.34	5.97	6.53
AU	Score 1-10 ↑	AirBench-chat	6.98	N/A	N/A	6.93	6.68	6.54
	ACC ↑	MMAU	55.56	N/A	N/A	52.50	61.23	53.29

Voxtral (Mistral, 2025)



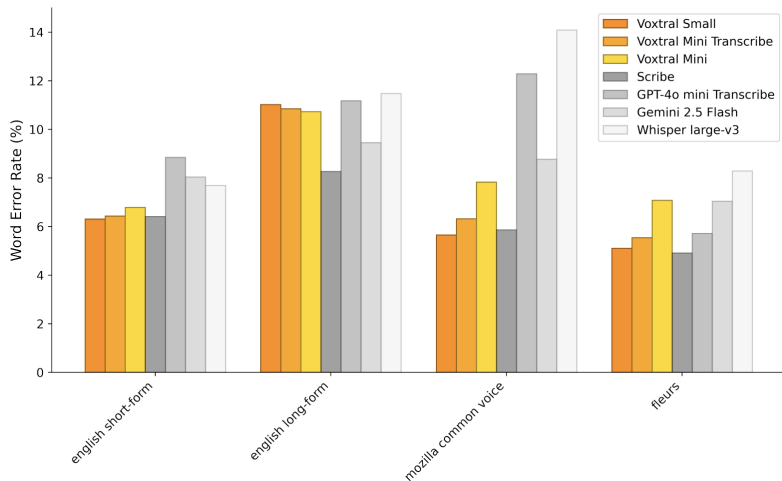
- ▶ Whisper Encoder + Linear Adapter + Transformer Decoder
- ▶ Subsampling: 4x to 12.5Hz
- ▶ Decoder: 3B for Mini, 24B for Small
- ▶ 3 stage train: pretrain, SFT, RL
- ▶ Online-DPO with text reward model, working on transcriptions + answers

Voxtral: pretrain data

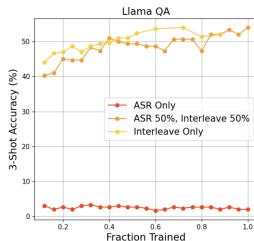
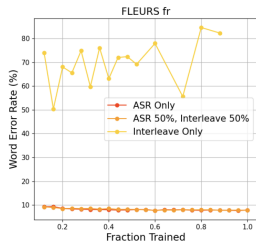
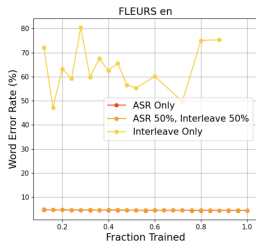


- ▶ Freeze all, except adapter
- ▶ Interleaving-style data
 1. Audio-text example (A, T) is first segmented into a set of audiotext pairs $(A_n, T_n)_N$
 2. **Audio-to-text repetition pattern:** a given audio A_n is repeated in the text space T_n
 3. **Cross-modal continuation pattern:** each audio A_n is followed by its subsequent text T_{n+1}

Voxtral: results



Voxtral: ablation on Interleaving-style data in pretrain



Voxtral: DPO effect

Model	% LLM Judge $\uparrow$	Grade $\uparrow$	En Short WER $\downarrow$
Voxtral Mini SFT	83.47 ± 2.17	3.92 ± 0.04	6.77
Voxtral Mini Offline DPO	84.91 ± 3.21	3.92 ± 0.08	6.78
Voxtral Mini Online DPO	85.59 ± 3.77	4.08 ± 0.07	6.79
Voxtral Small SFT	86.61 ± 0.96	4.16 ± 0.03	6.31
Voxtral Small Offline DPO	87.29 ± 1.65	4.19 ± 0.04	6.32
Voxtral Small Online DPO	88.31 ± 2.03	4.38 ± 0.06	6.50
GPT-4o mini Audio	80.00 ± 2.97	3.97 ± 0.05	-
Gemini 2.5 Flash	88.64 ± 2.28	4.54 ± 0.07	8.04

Audio representations for audio output

- ▶ Continuous generation is hard
 - ▶ $\Rightarrow$ the output is usually discrete audio tokens
 - ▶ Semantic tokens retain less prosodic information
 - ▶ $\Rightarrow$ acoustic tokens or a combination of both is beneficial
- $\Rightarrow$ Usually use codec models for speech decoder

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TTS leaderboard: LLM-based in the top

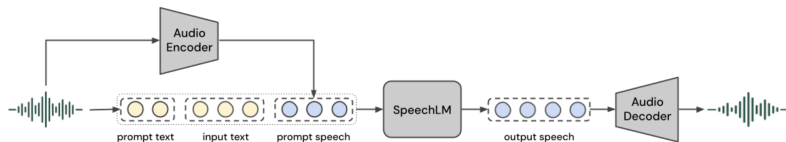
Speech Arena Leaderboard Artificial Analysis

Category: All		Accent: All		All	Open weights	Global Leaderboard	Personal Leaderboard	All models	Top models by creator
↑↓	Range	Creator	Model	ELO	95% CI	Samples	Released	API Pricing ¹	
1	1-2	Inworld	Inworld TTS 1.5 Max	1,215	-20/20	1,564	Jan 2026	\$10.0 /1M chars	
2	1-2	Google	Gemini 3.1 Flash TTS	1,211	-18/18	1,711	Apr 2026	\$36.6 /1M chars	
3	3-5	ElevenLabs	Eleven v3	1,179	-12/12	3,472	Feb 2026	\$171.9 /1M chars	
4	3-9	Inworld	Inworld TTS 1 Max	1,169	-20/20	1,380	Jun 2025	\$10.0 /1M chars	
5	3-7	MiniMax	Speech 2.8 HD	1,169	-12/12	3,215	Feb 2026	\$100.0 /1M chars	
6	3-11	Fish Audio	Fish Audio S2 Pro Open Weights	1,165	-33/33	486	Mar 2026	\$15.0 /1M chars	
7	3-9	Inworld	Inworld TTS 1.5 Mini	1,165	-17/17	1,821	Jan 2026	\$5.0 /1M chars	
8	8-9	MiniMax	Speech 2.8 Turbo	1,150	-12/12	3,349	Feb 2026	\$60.0 /1M chars	
9	4-12	StepFun	Step TTS 2 (Mar 2026)	1,150	-27/27	727	Feb 2026	\$50.0 /1M chars	
10	10-11	MiniMax	Speech 2.6 HD	1,135	-12/12	3,188	Oct 2025	\$100.0 /1M chars	

X-codec2 (from Llasa, 2025)

- ▶ Designed specifically for full architectural alignment with text-based LLMs (Llama)
- ▶ Semantic encoder: HuBERT → **Wav2Vec2-BERT**
- ▶ Quantization: RVQ → **Finite Scalar Quantization (FSQ)** (single-layer quantizer, no need to think about N tokens per step)
- ▶ Acoustic decoder: conv blocks → **Transformer + iSTFT**
- ▶ Codebook size: 1024 → **65536**

TTS-1 (Inworld, 2025)

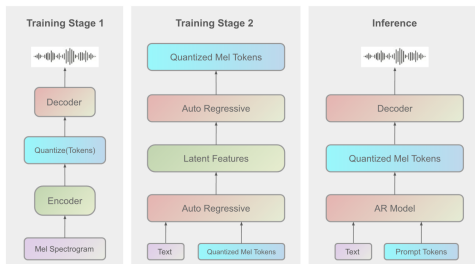


- ▶ Codec (X-codec2) + SpeechLM (LLaMA-3.2-1B, TTS-1-Max: LLaMA-3.1-8B)
- ▶ Why no diffusion/flow-matching: Streaming, harder to make RL, In-Context Learning better with LLMs
- ▶ 3 stages training:
 1. Pre-training (Next-token prediction): 1M hrs raw audio + 20B text tokens (Prevents text comprehension degradation)
 2. SFT: 200k hrs filtered audio-text

$[\langle \text{begin_text} \rangle, x_1, \dots, x_T, \langle \text{speech_start} \rangle, y_1, \dots, y_S, \langle \text{speech_end} \rangle]$

3. **RL-Alignment: GRPO** (from DeepSeek), Reward: Composite of WER, Speaker Similarity, DNSMOS

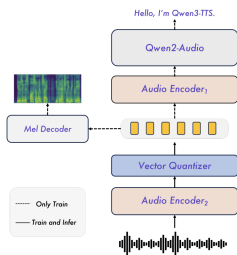
FishSpeech (FishAudio, 2024)



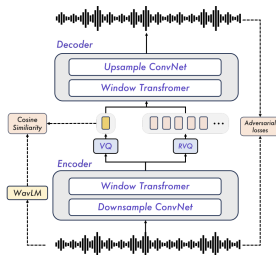
- ▶ Custom acoustic speech tokenizer with Grouped FSQ quantizer (RVQ analogue)
- ▶ 3 stages of training: pretraining, SFT, DPO
- ▶ 4B Llama-style backbone
- ▶ Dataset: 720k hours different langs (300 English, 300 Chinese, 120 others)

Atamanenko et al., Fish-Speech: Leveraging Large Language Models for Advanced Multilingual Text-to-Speech Synthesis, Fish Audio, 2024

Qwen3-TTS (Alibaba, 2026)



(a) Qwen-TTS-Tokenzier-25Hz



(b) Qwen-TTS-Tokenzier-12Hz

25Hz Variant:

- ▶ Single-codebook design.
- ▶ Semantic-acoustic hybrid features.
- ▶ Frame rate: **25 Hz** (40ms per token).

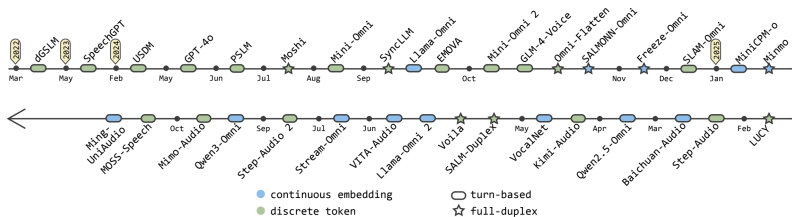
12Hz Variant:

- ▶ **16-layer** Multi-codebook RVQ.
- ▶ Semantic layer + 15 acoustic layers.
- ▶ Frame rate: **12.5 Hz** (80ms per token)

Outline

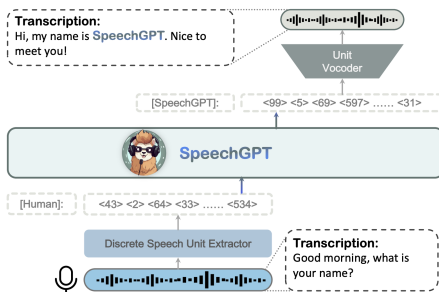
1. LLM-based audio models overview
2. Audio Conditioned LLMs: Seed-ASR, Llama3, Phi4, Voxtral
3. LLM-based TTS: TTS-1, FishSpeech, Qwen3-TTS
4. End-to-end speech interaction LLMs: SpeechGPT, Mini-Omni, Llama-Omni, Moshi, Qwen3-Omni

Timeline of end-to-end speech interaction LLMs



Wang et al., Towards General Auditory Intelligence: Large Multimodal Models for Machine Listening and Speaking, 2025

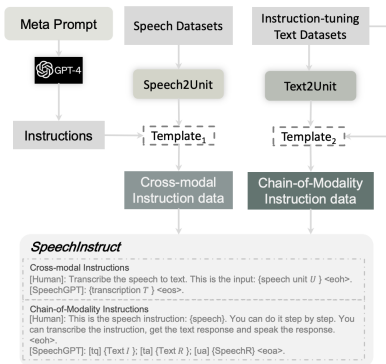
SpeechGPT (May, 2023)



- ▶ (Speech/Text)-to-(Speech/Text)
- ▶ LLM: **LLaMA**
- ▶ Discrete Unit Extractor: (**HuBERT**) + kMeans for audio tokenization
- ▶ Unit Vocoder (**HiFi-GAN**) (on tokens for token-to-audio)
- ▶ Data: GigaSpeech (ASR) + CommonVoice + LibriLight
- ▶ 3-stage training

Zhang et al., SpeechGPT: Empowering Large Language Models with Intrinsic Cross-Modal Conversational Abilities, Fudan University, 2023

SpeechGPT train stage 1: Modality-adaptation pre-training



Stage 1: Next-token prediction training with unlabeled speech corpus to enable LLM to handle discrete units modality

SpeechGPT train stage 2: Cross-modal instruction fine-tuning

Instruction: Can you transcribe the speech into a written format?

Input: **Speech clip** (Transcripts: I'm afraid there are no signs here said he.)

Output: **Text:** I'm afraid there are no signs here said he.

Instruction: Listen to the speech and write down its content.

Input: **Speech clip** (Transcripts: Did anyone know that these proofs would be there no one saved the printer.)

Output: **Text:** Did anyone know that these proofs would be there no one saved the printer.

Instruction: Would you mind speaking these words as naturally as possible?

Input: **Text:** Today is a sunny day and I'm happy to be here.

Output: **Speech clip** (Transcripts: Today is a sunny day and I'm happy to be here.)

Instruction: Would you please speed-read the following sentence?

Input: **Text:** I am a large language model that can listen and speak, a member of Fudan University, and glad to talk with you.

Output: **Speech clip** (Transcripts: I am a large language model that can listen and speak, a member of Fudan University, and glad to talk with you.)

1. Take ASR and TTS datasets with pairs of text-speech pairs
2. ChatGPT → create 100 instructions for ASR and TTS task
3. Assemble into instruction-text-speech or instruction-speech-text triplets
4. Train model on NTP task

Zhang et al., SpeechGPT: Empowering Large Language Models with Intrinsic Cross-Modal Conversational Abilities, Fudan University, 2023

SpeechGPT train stage 3: Chain-of-modality instruction fine-tuning

Speech Instruction-Speech Response:

[Human]: This is a speech instruction: [SpeechI]. And your response should be speech. You can do it step by step. You can first transcribe the instruction and get the text instruction. Then you can think about the instruction and get the text response. Last, you should speak the response aloud <eoh>. [SpeechGPT]: [tq] {TextI}; [ta] {TextR}; [ua] {SpeechR}<eoa>.

Speech Instruction-Text Response:

[Human]: This is a speech instruction: [SpeechI]. And your response should be text. You can do it step by step. You can first transcribe the instruction and get the text instruction. Then you can think about the instruction and get the text response. <eoh>. [SpeechGPT]: [tq] {TextI}; [ta] {TextR}<eoa>.

Text Instruction-Speech Response:

[Human]: This is a text instruction: [TextI]. And your response should be speech. You can do it step by step. You can think about the instruction and get the text response. Then you should speak the response aloud <eoh>. [SpeechGPT]: [ta] {TextR}; [ua] {SpeechR}<eoa>.

Text Instruction-Text Response:

[Human]: This is a text instruction: [TextI]. And your response should be text. You can think about the instruction and get the text response. [SpeechGPT]: [ta] {TextR}<eoa>.

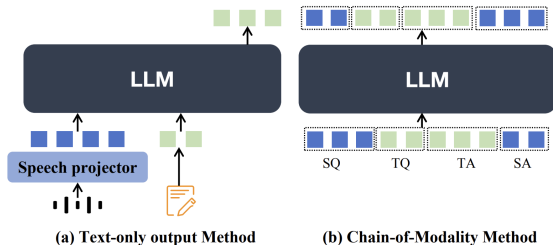
1. Train a model, which generates HuBERT tokens from text (Encoder-Decoder transformer, trained on LibriSpeech)
2. Text dialogue dataset $\rightarrow$ above model $\rightarrow$ generate quadruplets for each sample in a dialogue dataset:

$(TextI, TextR) \rightarrow (SpeechI, TextI, TextR, SpeechR)$

3. Dataset pair (speech input + speech output) $\rightarrow$ Fine tune model with PEFT (LoRA) for further modality alignment

Zhang et al., SpeechGPT: Empowering Large Language Models with Intrinsic Cross-Modal Conversational Abilities, Fudan University, 2023

Generation paradigms: Chain-of-Modality Method

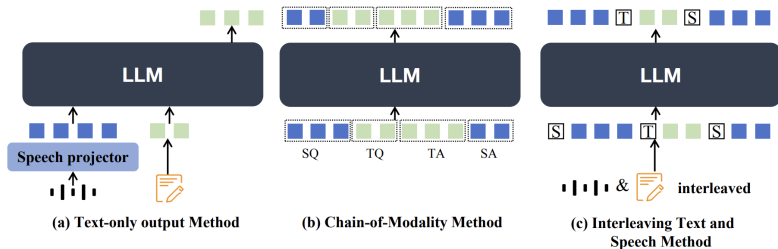


First generates response text autoregressively before producing the corresponding speech (this enhance the quality of the response content), but **it is not suitable for live interactions**

1. SpeechGPT (2023)
2. Spectron (Google Research, 2023)
3. EMOVA (2024)

Do we really need an ASR stage?

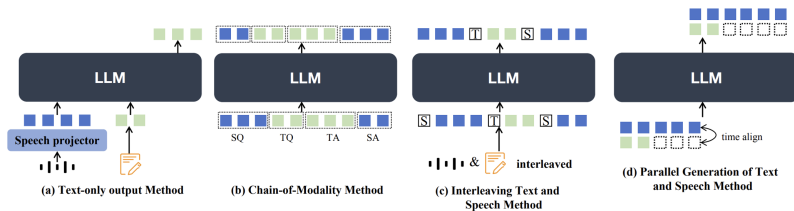
Generation paradigms: Interleaving Text and Speech



1. Spirit LM (Meta AI, 2024)
2. USDM (NAVER, 2024)
3. GLM-4-Voice (Zhipu AI, 2024)
4. Step-Audio 2 (2025)

How can we further reduce latency?

Generation paradigms: Parallel Generation of Text and Speech



1. **PSLM** (2024)
2. **Mini-Omni** (Inspir AI, 2024)
3. **Moshi** (Kyut AI, 2024)
4. **Mini-Omni2** (Inspir AI, 2024)

Mini-Omni: opensource GPT4o analogue

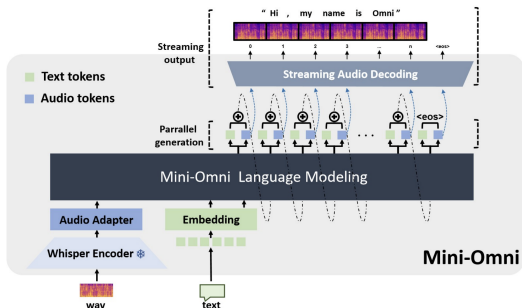


Figure: The first fully end-to-end, open-source model for real-time speech interaction

- ▶ **LLM** backbone: Qwen2-0.5B
- ▶ **Audio encoding:** Whisper-small; **decoding:** SNAC Codec
- ▶ **ASR Adapter:** 2-layer MLP; **TTS Adapter:** 6 transformer blocks

Xie et al., *Mini-Omni: Language Models Can Hear, Talk While Thinking in Streaming*, 2024

Mini-Omni training stages

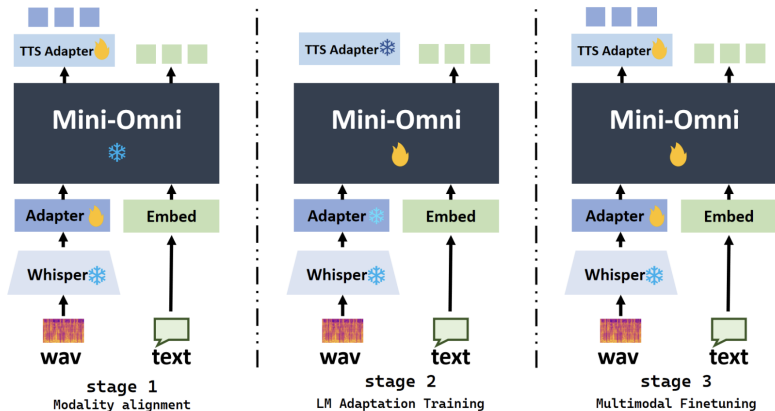


Figure: Three-stage training phases: modality expansion, modality adaptation training, holistic fine-tuning

Llama-Omni

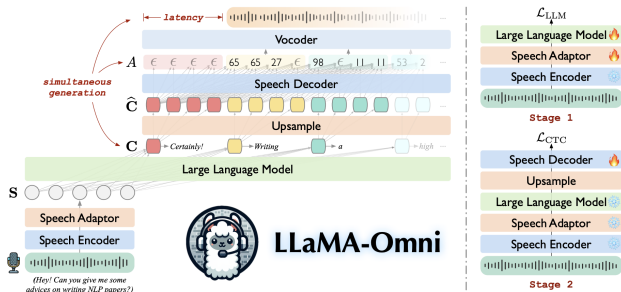
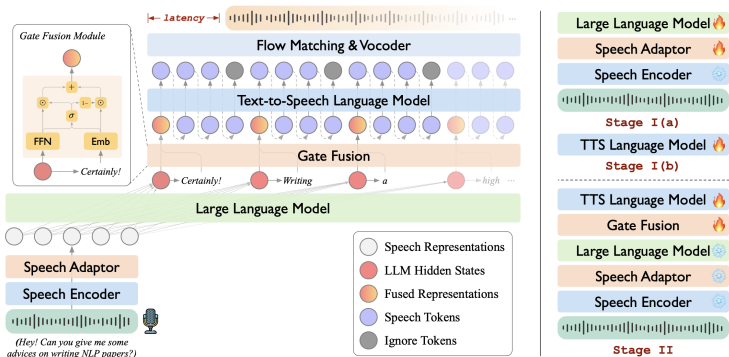


Figure: **Left:** Model architecture **Right:** Two-stage training strategy

- ▶ **Whisper-large-v3** + 5x downsampling + **Llama-3.1-8B-Instruct** + 2 LLaMA Transformer layers
- ▶ Non-Autoregressive (NAR) CTC Streaming Decoder: **HuBERT** tokens + **HiFi-GAN** vocoder with a duration predictor

Fang et al., LLaMA-Omni: Seamless Speech Interaction with Large Language Models, ICLR, 2025

Llama-Omni 2



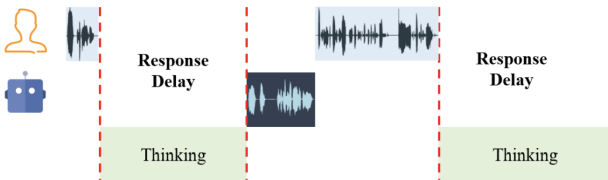
Change TTS decoder: NAR $\rightarrow$ AR (add Gate Fusion, Chunk-aware Causal Flow Matching from CosyVoice 2)

Fang et al., LLaMA-Omni2: LLM-based Real-time Spoken Chatbot with Autoregressive Streaming Speech Synthesis, 2025

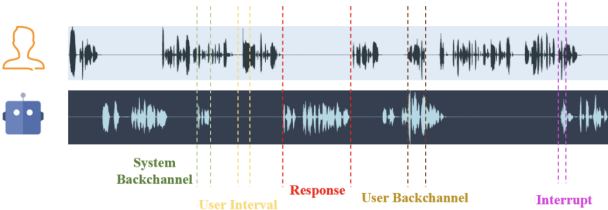
Types of interactions



(a) Simplex: One-way communication, and the direction is fixed.

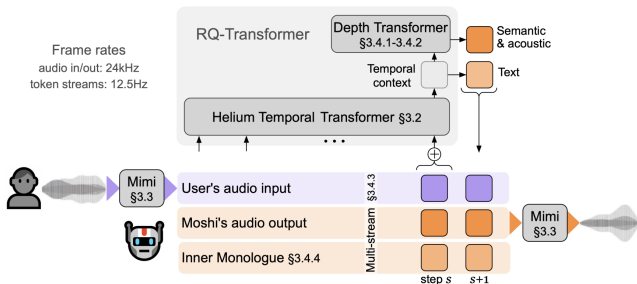


(b) Half-duplex: Two-way communication, but not simultaneously.



(c) Full-duplex: Two-way communication, simultaneously.

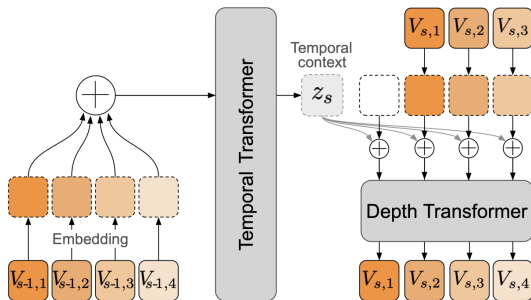
Moshi (Kyutai AI, 2024)



- ▶ Model listens and speaks simultaneously (full duplex)
- ▶ **Helium**: own LLM with 7B parameters, trained on 2.1T tokens (Llama2 - 1.8T tokens, Llama3 - 15.6T tokens)
- ▶ **Mimi**: Neural RVQ codec, semantic+acoustic tokens
- ▶ **Moshi**: Fine-tuned Helium to predict audio tokens
- ▶ **Inner monologue**: Training and inference setup for audio language models

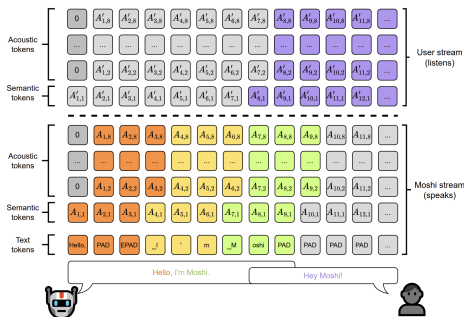
Défossez et al., *Moshi: a speech-text foundation model for real-time dialogue*, Kyutai AI Lab, 2024

Moshi: RQ transformer



- ▶ Add an small autoregressive transformer over a pretrained backbone
- ▶ Flattened sequence of length $K \times S$ into S timesteps for a large Temporal Transformer which produces a context embedding used to condition a smaller Depth Transformer over K steps

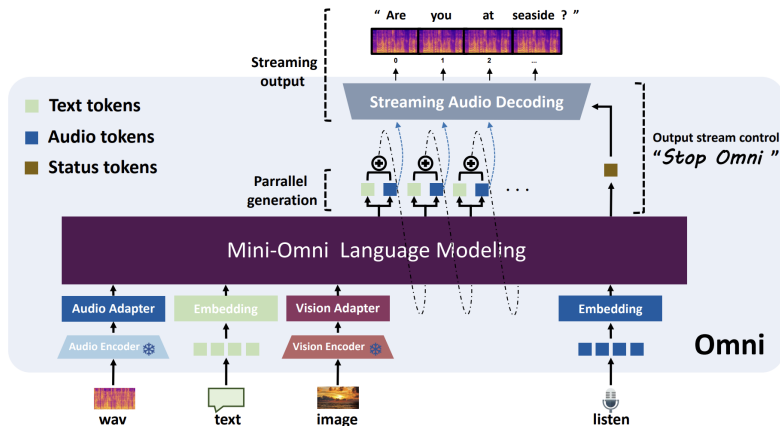
Moshi full duplex interaction: Inner monologue



- ▶ Stream of users speech: 8 layers (1 semantic and 7 acoustic)
- ▶ Stream of synthesized speech: of 8 layers (1 semantic and 7 acoustic)
- ▶ Acoustic streams are delayed, comparing to semantic (model could see 1 semantic token ahead)
- ▶ 1 text token (thinking) $\rightarrow 8 + 8 + 1 = 17$

Défossez et al., *Moshi: a speech-text foundation model for real-time dialogue*, Kyutai AI Lab, 2024

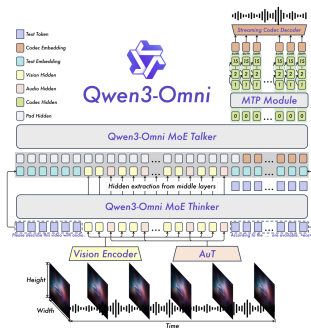
Mini-Omni 2 (2024)



- ▶ Add visual modality
- ▶ Add full-duplex idea (reacts to phrase "Stop Omni")

Zhifei Xie, Changqiao Wu, *Mini-Omni2: Towards Open-source GPT-4o with Vision, Speech and Duplex Capabilities*, Inspirai, 2024

Qwen3-Omni (2025)



- ▶ **Thinker-talker paradigm** (from Qwen2.5-omni): Thinker is tasked with text generation while the Talker focuses on generating streaming speech tokens
- ▶ Use MoE models instead of Dense
- ▶ Inspired by the way humans utilize different organs to produce various signals, which are simultaneously coordinated through the same neural networks

Speech Reasoning (Big Bench Audio) vs. Speed

